Given:

$$X \sim N(2.1,\ 0.3)$$

Mean:

$$\mu = 2.1$$

Standard deviation:

$$\sigma = 0.3$$

---

**a) Probability between 2.0 g/m and 2.5 g/m**

$$P(2.0 < X < 2.5)$$

Find Z-scores:

$$Z_1 = \frac{2.0 - 2.1}{0.3} = -0.33$$
$$Z_2 = \frac{2.5 - 2.1}{0.3} = 1.33$$

So:

$$P(2.0 < X < 2.5) = P(-0.33 < Z < 1.33)$$

From Z-table:

$$P(Z < 1.33) = 0.9082$$
$$P(Z < -0.33) = 0.3707$$
$$P(-0.33 < Z < 1.33) = 0.9082 - 0.3707$$
$$= 0.5375$$

**Answer:**

$$\boxed{0.5375}$$

or **53.75%**

---

**b) Proportion in violation of 2.7 g/m mandate**

Violation means emissions exceed 2.7:

$$P(X > 2.7)$$

Find Z-score:

$$Z = \frac{2.7 - 2.1}{0.3} = 2.00$$

From Z-table:

$$P(Z < 2.00) = 0.9772$$
$$P(Z > 2.00) = 1 - 0.9772$$
$$= 0.0228$$

**Answer:**

$$\boxed{0.0228}$$

or **2.28%**

---

**c) Score corresponding to the top 10%**

Top 10% means:

$$P(X > x) = 0.10$$

So:

$$P(X < x) = 0.90$$

From Z-table:

$$Z = 1.28$$

Use:

$$X = \mu + Z\sigma$$
$$X = 2.1 + (1.28)(0.3)$$
$$X = 2.1 + 0.384$$
$$X = 2.484$$

**Answer:**

$$\boxed{2.48 \text{ g/m}}$$

---

**d) New mean so that at most 5% exceed 2.7 g/m**

We want:

$$P(X > 2.7) = 0.05$$

So:

$$P(X < 2.7) = 0.95$$

From Z-table:

$$Z = 1.645$$

Use:

$$Z = \frac{X - \mu}{\sigma}$$
$$1.645 = \frac{2.7 - \mu}{0.3}$$
$$1.645(0.3) = 2.7 - \mu$$
$$0.4935 = 2.7 - \mu$$
$$\mu = 2.7 - 0.4935$$
$$\mu = 2.2065$$

**Answer:**

$$\boxed{2.21 \text{ g/m}}$$

The mean must be reduced to approximately **2.21 g/m**.